

Technology Stack

Date	11 July 2026
Team ID	xxxxxx
Project Name	StudyAI – Smart AI Study Companion & Quiz Generator
Maximum Marks	2 Marks

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, Vite, Tailwind CSS, Framer Motion	Provides a fast, responsive, and modern user interface with reusable components, efficient state management, responsive design, and smooth animations for an enhanced user experience.
2	Backend / Server-Side	Python, Flask, Gunicorn	Flask offers a lightweight and modular backend architecture for REST APIs, while Gunicorn enables efficient production deployment and scalability.
3	Database / Data Storage	Firebase Firestore, Local JSON Storage (Fallback)	Firestore provides secure cloud-based NoSQL storage with real-time synchronization, while local JSON storage ensures uninterrupted functionality when

			Firestore is unavailable.
4	Cloud / Hosting / Deployment	Vercel, Render, Firebase	Vercel hosts the React frontend with automatic deployments, Render hosts the Flask backend, and Firebase provides cloud database services for scalable and reliable deployment.
5	Version Control & CI/CD	Git, GitHub	Git enables version control and collaboration, while GitHub provides repository hosting, source code management, and automated deployment integration.
6	Third-Party APIs / Other Tools	Groq API (Llama 3.3 70B Versatile), Firebase Authentication, Chart.js, React Markdown, Axios	Groq API powers AI-based summaries, flashcards, quizzes, and study plans; Firebase Authentication manages secure user login; Chart.js visualizes learning analytics; React Markdown renders AI-generated content; Axios enables efficient frontend-backend communication.